Tessera Badge

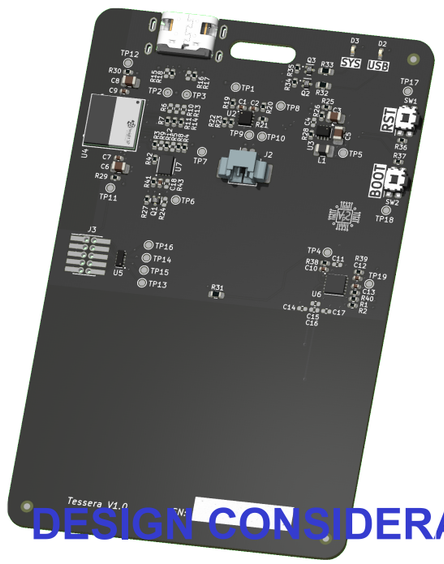
Variant: PRELIMINARY

2026-10-02

Rev + (Unreleased)

Page	Index	Page	Index	Page	Index	Page	Index
1	Cover Page	11	.....	21	.....	31	.....
2	Block Diagram	12	.....	22	.....	32	.....
3	Project Architecture	13	.....	23	.....	33	.....
4	USB	14	.....	24	.....	34	.....
5	power	15	.....	25	.....	35	.....
6	MCU	16	.....	26	.....	36	.....
7	Leds	17	.....	27	.....	37	.....
8	NFC	18	.....	28	.....	38	.....
9	Secure element	19	.....	29	.....	39	.....
10	Revision History	20	.....	30	.....	40	.....

TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as ✗

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Date: 02-Oct-2026

DESIGN CONSIDERATIONS

DESIGN NOTE: Example text for informational design notes.

DESIGN NOTE: Example text for debug notes.

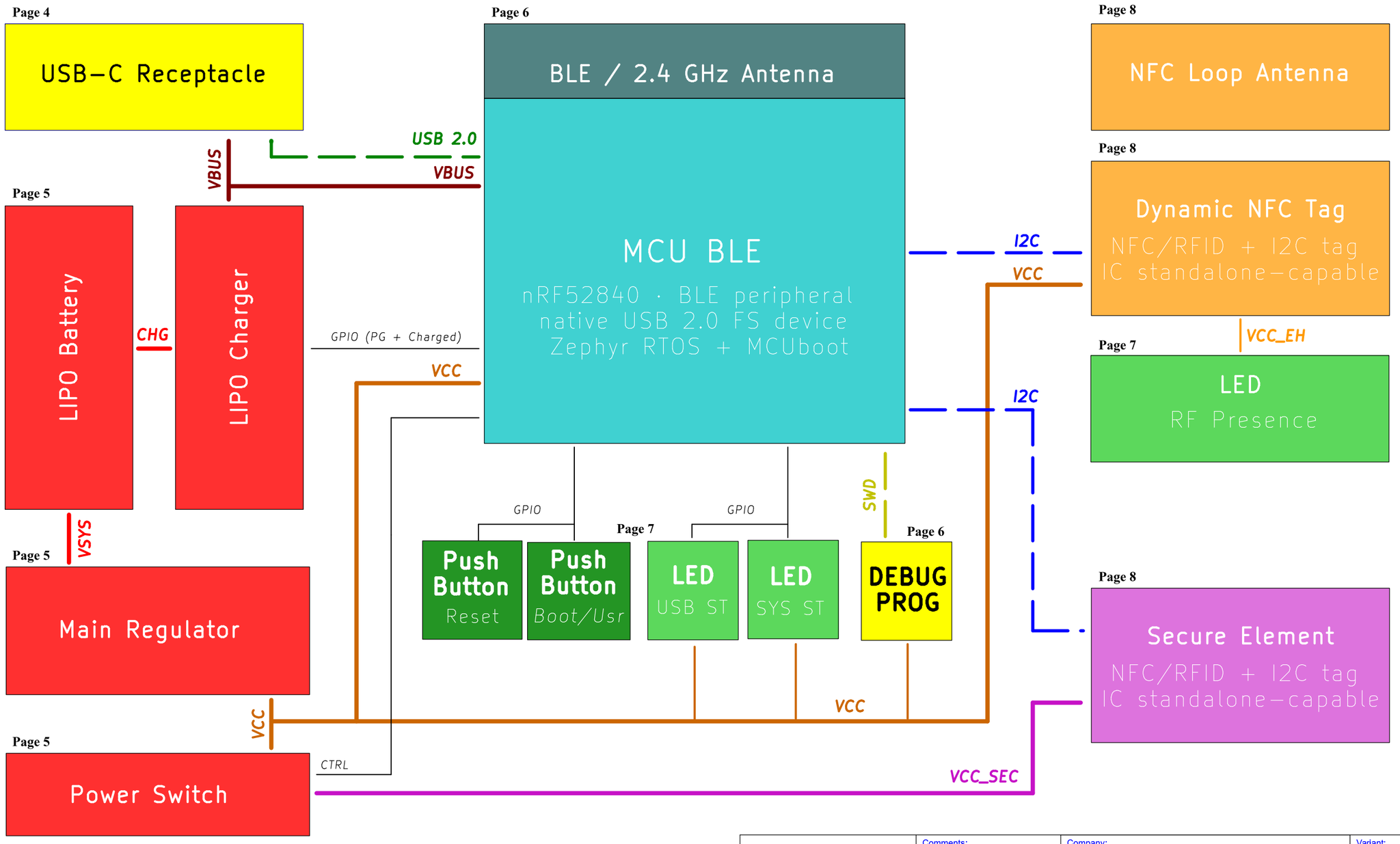
DESIGN NOTE: Example text for cautionary design notes.

DESIGN NOTE: Example text for critical design notes.

LAYOUT NOTE: Example text for critical layout guidelines.

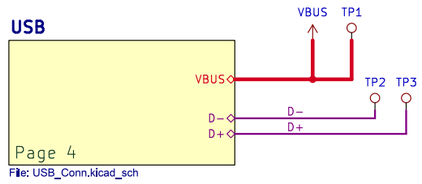
Comments:	Company: APS		Variant: PRELIMINARY	Git Hash: 3a208dc
	Board Name: Tessera Badge		Project Name: Tessera	
Sheet Title: Front Page	File Name: tessera.kicad_sch	Designer: Antonio Morales	Date: 2026-10-01	Revision: + (Unreleased)
Sheet Path: /		Reviewer:	Size: A3	Sheet: 1 of 10

[2] Block Diagram

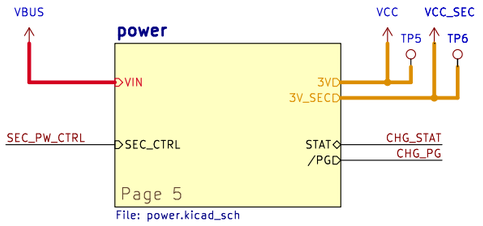


Comments:	Company: APS		Variant: PRELIMINARY	Git Hash: 3a208dc
	Board Name: **Tessera Badge**		Project Name: **Tessera**	
Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Antonio Morales	Date: 2026-10-01	Revision: + (Unreleased)
Sheet Path: /Block Diagram/		Reviewer:	Size: **A3**	Sheet: **2** of **10**

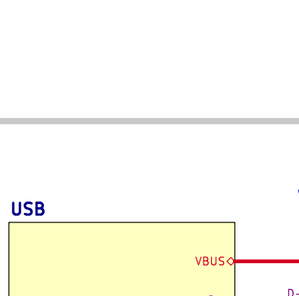
[3] Project Architecture



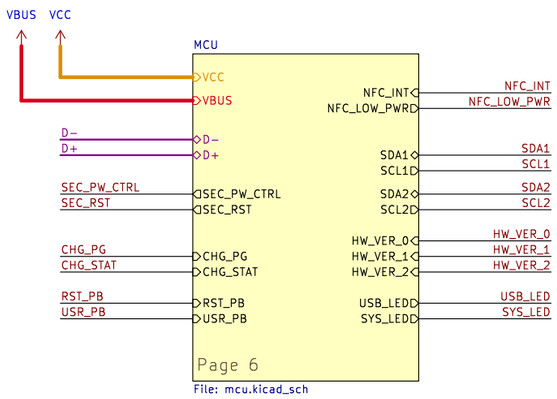
External power supply & USB connection



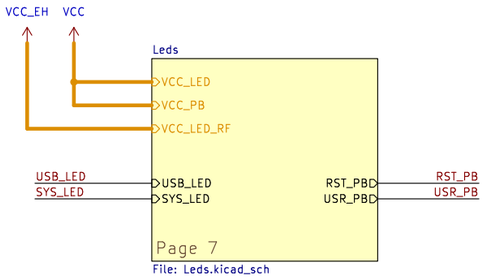
Batt charger & Main power supply



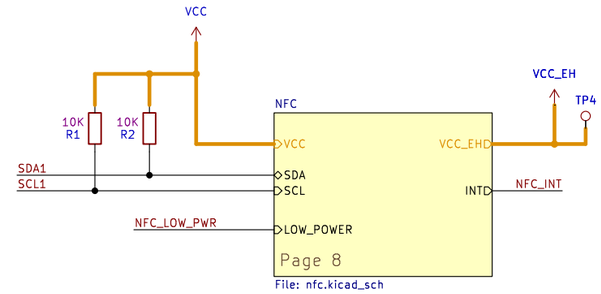
Board Hardware Version Selection 000b



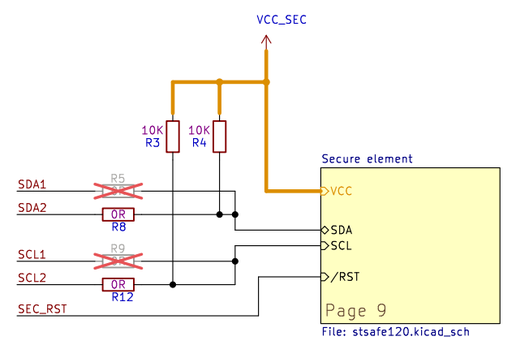
MCU & BLE Radio



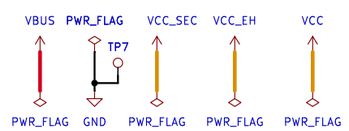
User IO: Leds & Push Button



NFC Tag & Antenna



Secure Element Processor & Storage



ERC Power Flags



Fiducials

G1
WA_APS_Logo

G2
WA_tessera_Name

G3
WA_tessera_light_bulb

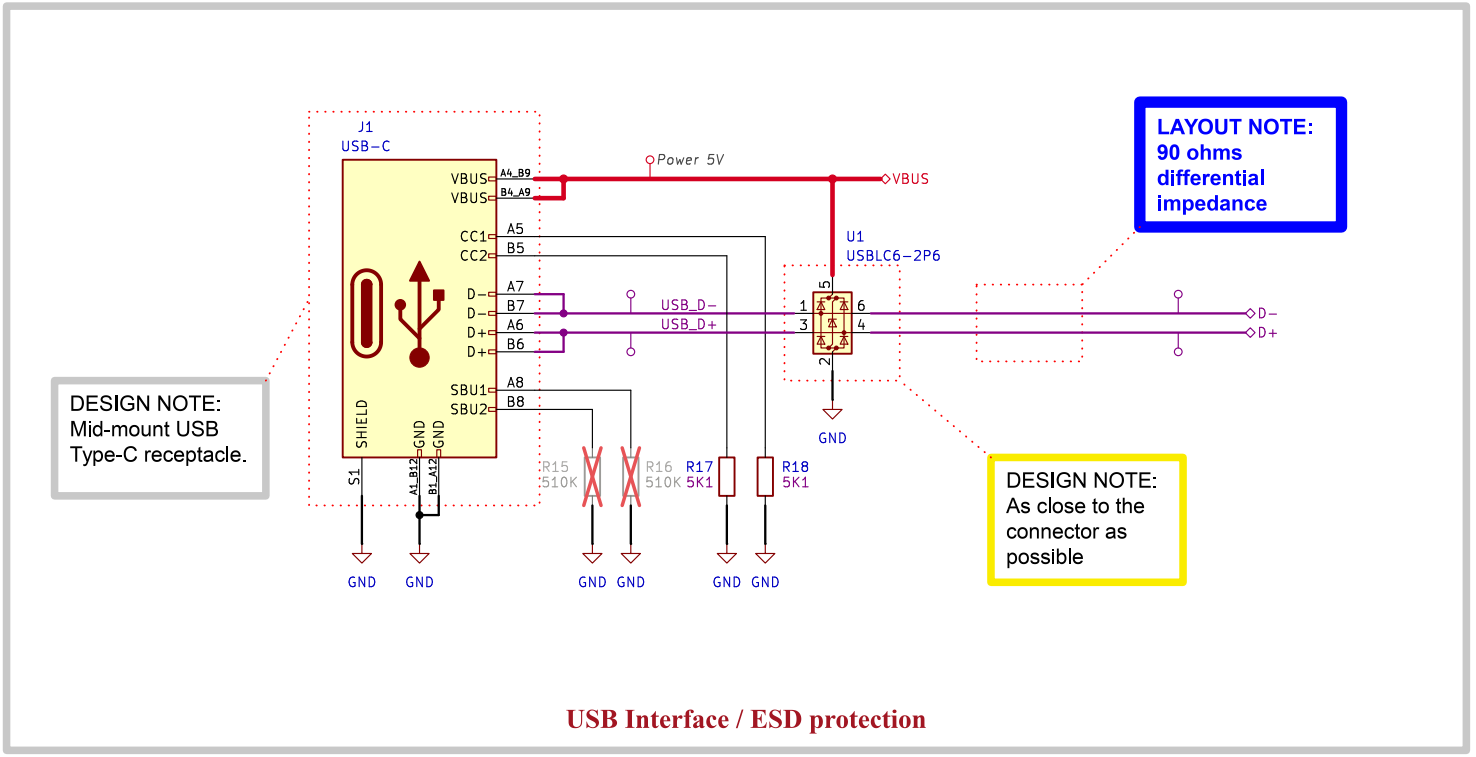
G4
WA_tessera_linkedin_qr

G5
WA_tessera_binary

Workart

Comments:	Company: APS		Variant: PRELIMINARY	Git Hash: 3a208dc
	Board Name: **Tessera Badge**		Project Name: **Tessera**	
Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Antonio Morales	Date: 2026-10-01	Revision: + (Unreleased)
Sheet Path: /Project Architecture/		Reviewer:	Size: **A3**	Sheet: **3** of **10**

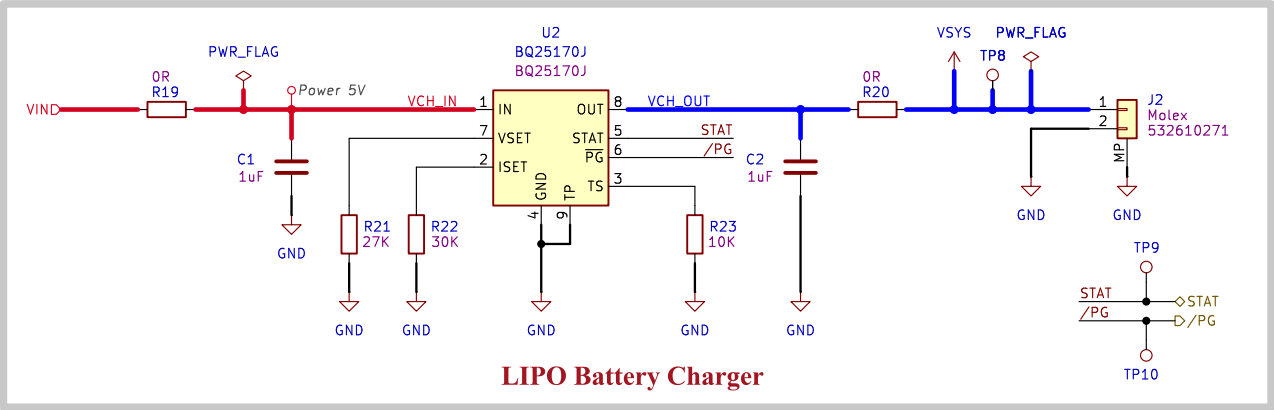
[4] USB Connector



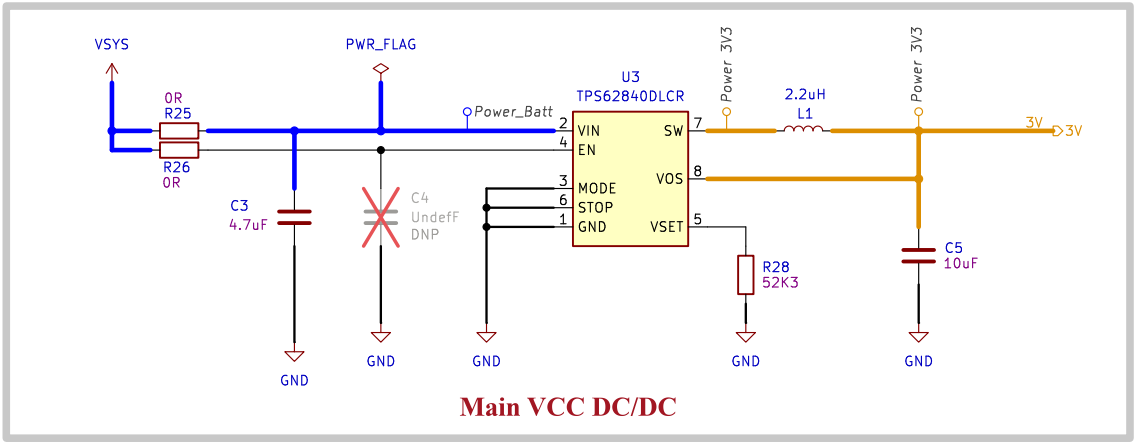
USB Interface / ESD protection

Comments:	Company: APS		Variant: PRELIMINARY	Git Hash: 3a208dc
	Board Name: Tessera Badge		Project Name: Tessera	
Sheet Title: USB Connector	File Name: USB_Conn.kicad_sch	Designer: Antonio Morales	Date: 2026-10-01	Revision: + (Unreleased)
Sheet Path: /Project Architecture/USB/		Reviewer:	Size: A4	Sheet: 4 of 10

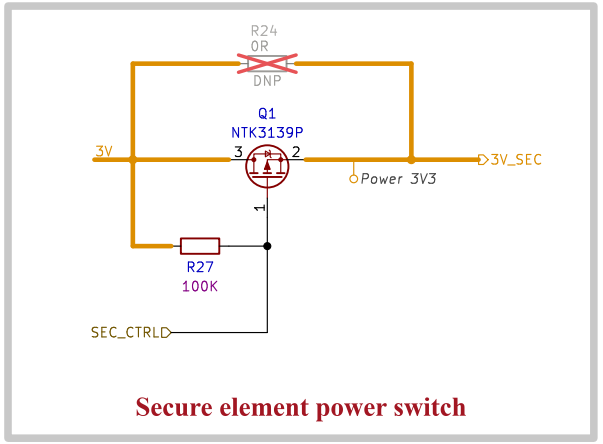
[5] Power Supply



LIPO Battery Charger



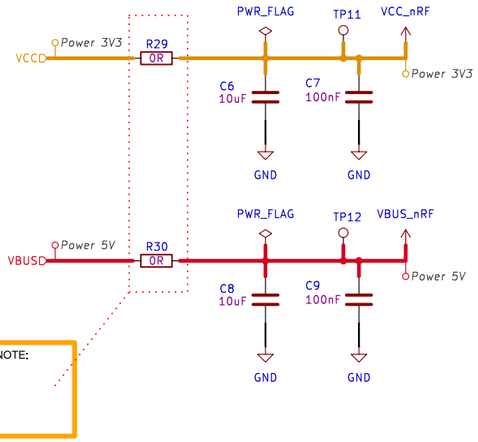
Main VCC DC/DC



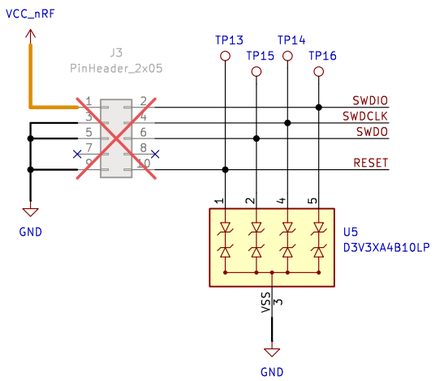
Secure element power switch

Comments:	Company:		Variant: PRELIMINARY	Git Hash: 3a208dc
	Board Name: **Tessera Badge**		Project Name: **Tessera**	
Sheet Title: Power Supply	File Name: power.kicad_sch	Designer: Antonio Morales	Date: 2026-10-01	Revision:
Sheet Path: /Project Architecture/power/		Reviewer:	Size: **A4**	Sheet: **5** of **10**

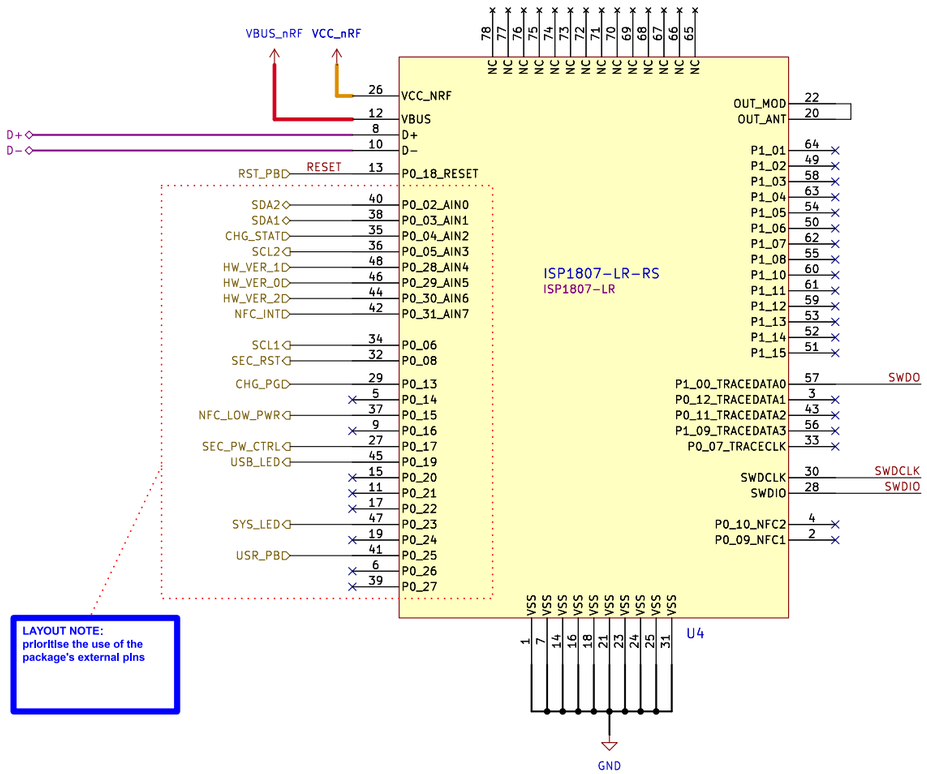
[6] BLE MCU



MCU Power supply



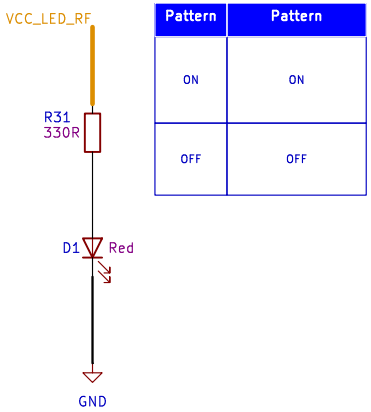
Debug & Program interface



Main MCU

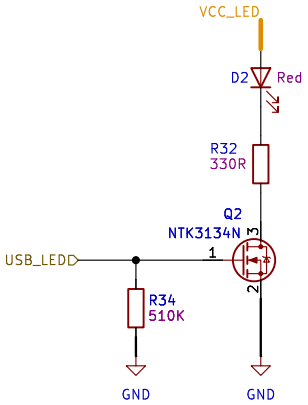
Comments:	Company:		Variant: PRELIMINARY	Git Hash: 3a208dc
	Board Name: **Tessera Badge**		Project Name: **Tessera**	
Sheet Title: BLE MCU	File Name: mcu.kicad_sch	Designer: Antonio Morales	Date: 2026-10-01	Revision:
Sheet Path: /Project Architecture/MCU/		Reviewer:	Size: **A3**	Sheet: **6** of **10**

[7] User IO



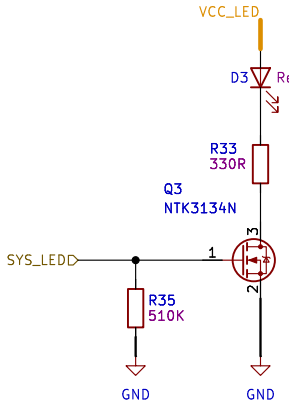
Pattern	Pattern
ON	ON
OFF	OFF

NFC activity LED



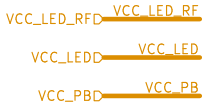
Pattern	State
ON	Host has enumerated and mounted/opened the device (USB configured, crypto-token session open where applicable). Do not unplug.
OFF	Not on USB power, not yet enumerated, or session cleanly closed. Safe to unplug.
Blinking	

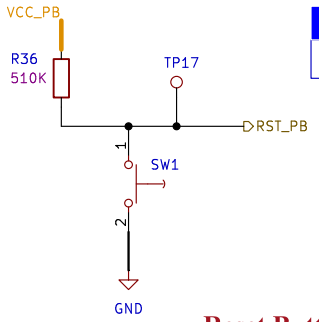
USB Status LED



Pattern	State
500 ms ON, 1500 ms OFF, repeating	Bootloader / DFU active
3× (50 ms ON, 200 ms OFF), one-shot	Application start
5× (50 ms ON, 50 ms OFF), one-shot per event	NFC event while powered
200 ms ON, 200 ms OFF, repeating	Crypto operation in progress
20 ms ON every 2000 ms	BLE advertising
2x 20 ms ON every 2000 ms	BLE connected
1000 ms ON, 1000 ms OFF, repeating	USB charging in progress
Solid ON	USB charging complete
100 ms ON, 100 ms OFF, continuous	Error
OFF	Idle

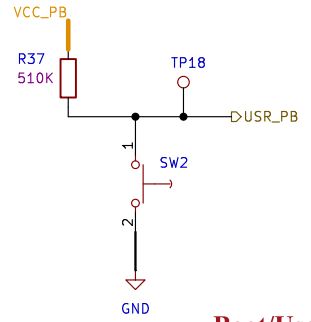
System Status LED





Action	Result
Push	Reset the device

Reset Button

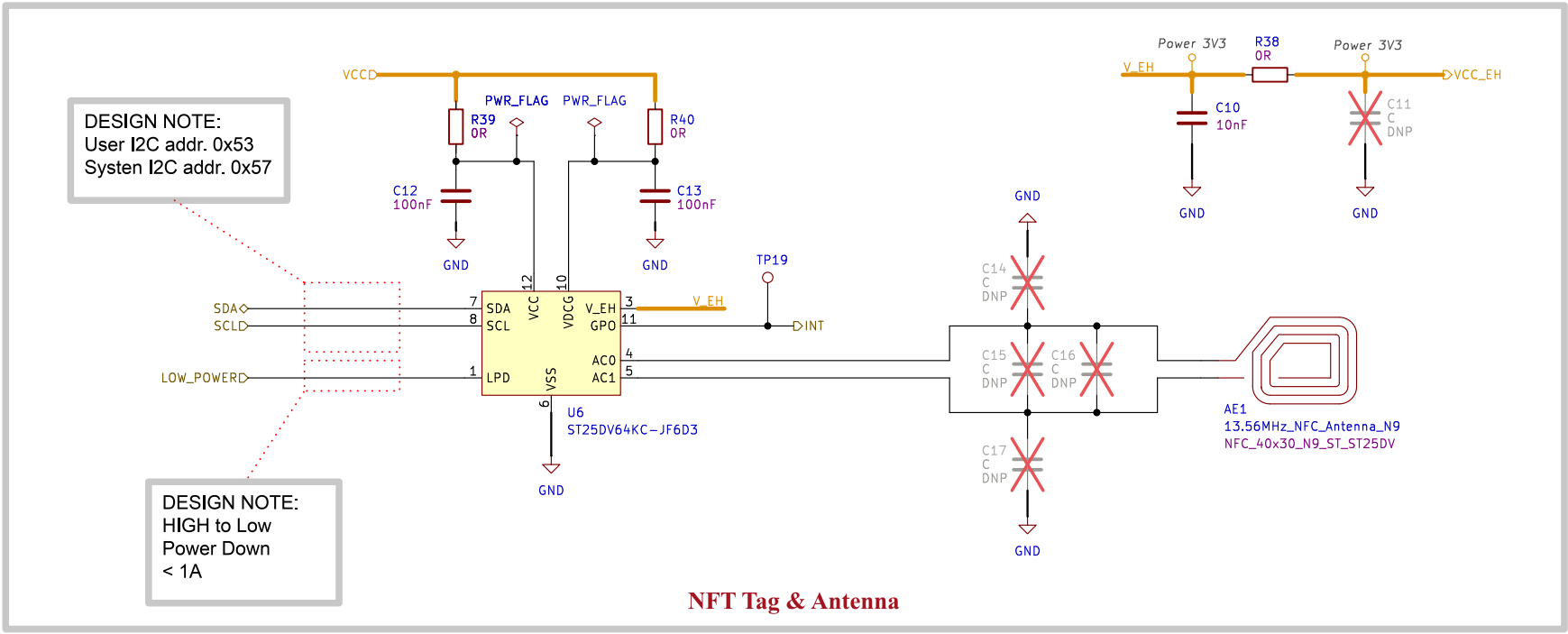


Action	Result
Maintaid Pushed while booting	Stay in bootloader for DFU operation
Short push	Action 1
Short push	Action 1
Short push	Action 1

Boot/User Button

Comments:	Company:		Variant: PRELIMINARY	Git Hash: 3a208dc
	Board Name: **Tessera Badge**		Project Name: **Tessera**	
Sheet Title: User IO	File Name: Leds.kicad_sch	Designer: Antonio Morales	Date: 2026-10-01	Revision:
Sheet Path: /Project Architecture/Leds/		Reviewer:	Size: **A4**	Sheet: **7** of **10**

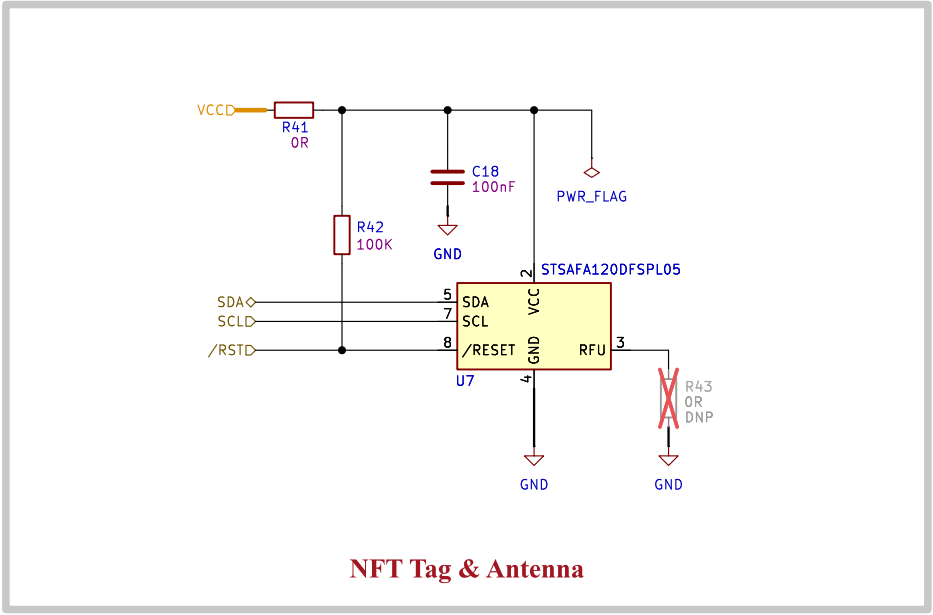
[8] NFC Interface



Signal Name	Function	Direction
V_EH	Supply voltage for GPO driver	Power output
GPO	Interrupt output	Output
SDA	Serial data	I/O
SCL	Serial clock	Input
AC0, AC1	Antenna coils	–
VCC	Supply voltage	Power
VSS	Ground	–
LPD	Low power down mode	Input
VDCG	Supply voltage for GPO driver	Power output
NC	Not connected	–
EP	Exposed pad	Must be left floating

Comments:	Company:		Variant: PRELIMINARY	Git Hash: 3a208dc
	Board Name: **Tessera Badge**		Project Name: **Tessera**	
Sheet Title: NFC Interface	File Name: nfc.kicad_sch	Designer: Antonio Morales	Date: 2026-10-01	Revision:
Sheet Path: /Project Architecture/NFC/		Reviewer:	Size: **A4**	Sheet: **8** of **10**

[9] Secure Element



NFT Tag & Antenna

Comments:	Company:		Variant: PRELIMINARY	Git Hash: 3a208dc
	Board Name: **Tessera Badge**		Project Name: **Tessera**	
Sheet Title: Secure Element	File Name: stsafe120.kicad_sch	Designer: Antonio Morales	Date: 2026-10-01	Revision:
Sheet Path: /Project Architecture/Secure element/		Reviewer:	Size: **A4**	Sheet: **9** of **10**

[10] Revision History

Comments:	Company: APS		Variant: PRELIMINARY	Git Hash: 3a208dc
	Board Name: **Tessera Badge**		Project Name: **Tessera**	
Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Antonio Morales	Date: 2026-10-01	Revision: + (Unreleased)
Sheet Path: /Revision History/		Reviewer:	Size: **A4**	Sheet: **10** of **10**